LEARN / DATA CENTERS / BEGINNER EDITION

POWER & COOLING

Every computation becomes heat. Reliability begins before the code runs.

SCAN / LEARN

THE SYSTEM AT A GLANCE



FOUR IDEAS THAT MATTER

1 ELECTRICAL PATH

Power reaches the facility, passes through electrical distribution and supplies computers.

2 UPS (UNINTERRUPTIBLE POWER SUPPLY)

A short-term power system that can bridge interruptions while other protection acts.

3 HEAT HAS TO MOVE

Fans or liquid cooling carry heat away. More computation generally requires more heat removal.

4 EFFICIENCY IS END TO END

Useful work per unit of energy depends on hardware, cooling, utilization and software decisions.

MEMORY HOOK

A server room is a machine whose purpose is to keep other machines usable.

DISCUSS WITHOUT COMPUTERS

What happens to a busy AI cluster if cooling stops?

© 2026 VLAD BELCIUG • FREE FOR PERSONAL STUDY